

Owade Kennedy Jackob

Curriculum Vitae

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Personal Information

Nationality Kenyan
Gender Male
Languages English, Swahili, Luo

Expertise & Research Interests

- **Mathematical Modeling** (Environmental Systems and Epidemiology for predictions and policy decisions)
- **Differential Equations** (ODEs/PDEs), Dynamical Systems, Numerical Analysis and Fluid Mechanics
- **Computational Tools**: MATLAB, Python, and R for Data Analysis
- **Technical Writing**: $\text{\LaTeX}$ for professional typesetting

Professional Membership

2024–Present **Member**, *European Mathematical Society*, Helsinki, Finland
2017–Present **Member**, *Teachers Service Commission*, Nairobi, Kenya

Education

Feb 2019–Dec 2024 **Doctor of Philosophy in Applied Mathematics**, *Masinde Muliro University of Science and Technology*, Kakamega, Kenya
Thesis: Mathematical Model for Cholera Dynamics with Vaccination and Temperature-Dependent Parameters
Supervisors: Dr. Akinyi Colleta, Dr. Frankline Tireito

2015–Dec 2018 **Master of Science in Applied Mathematics**, *Masinde Muliro University of Science and Technology*, Kakamega, Kenya
Thesis: Modeling the Role of Rehydration and Antibiotic Treatment on Reduction of Cholera Mortality
Supervisors: Dr. Akinyi Colleta, Prof. George Lawi
Courses: Partial Differential Equations I/II, Ordinary Differential Equations I/II, Numerical Analysis I/II, Advanced Classical Analysis, Applied Dynamical Systems I/II, Biomathematics I/II, Complex Analysis I, Numerical Computation of DEs and Stochastic Differential Equations

2010–July 2014 **Bachelor of Education Science (Mathematics and Physics)**, *Masinde Muliro University of Science and Technology*, Kakamega, Kenya
Second Class Honours, Upper Division

2004–2008 **Kenya Certificate of Secondary Education (KCSE)**, *Kangeso Boys High School*, Rongo - Migori County, Kenya

1995–2003 **Kenya Certificate of Primary Education (KCPE)**, *Rabwao Primary School*, Muhuru-Bay, Kenya

Work Experience

2016–Present **Part-time Lecturer of Mathematics, Masinde Muliro University of Science and Technology, Kakamega, Kenya**

Responsibilities:

- Teach and supervise undergraduate and postgraduate students in the department of Mathematics
- Courses taught:
 - MAT 321 – Ordinary Differential Equations I
 - MAT 428 – Mathematical Modelling
 - MAT 122 – Elementary Applied Mathematics
 - MAT 422 – Partial Differential Equations II
 - MAT 203 – Vector Analysis
 - MAT 422 – Ordinary Differential Equation II
 - MAT 102 – Analytic Geometry
 - MAT 201E – Engineering Mathematics I
 - MAT 202E – Engineering Mathematics II
 - MAT 301E/MAT 361 – Engineering Mathematics III/Fourier Series and Laplace Transform
 - MAT 302E/MAT 362 – Engineering Mathematics IV/Differential Equations
 - MAT 121 – Differential Calculus
 - MAT 124/221 – Integral Calculus
 - MAT 423 – Ordinary Differential Equations II
 - MAT 162 – Pure Mathematics II
 - MAT 426 – Fourier Series
 - MAT 851 – Ordinary Differential Equations I (Masters class)
- Participate in setting, moderating, marking and analyzing examinations for undergraduate courses
- Supervise undergraduate students in Applied Mathematics projects
- B.Sc. and M.Sc. Mathematics students' academic advisor
- Masinde Muliro University (MMUST) ODeL instructor (Online Distance and E-learning - 2015/2016 Academic Year)

2019–Present **Part-time Lecturer of Mathematics, Kaimosi Friends University, Kaimosi, Kenya**

Responsibilities:

- Teach undergraduate students in the department of Mathematics
- Courses taught:
 - MAT 428 – Mathematical Modeling
 - MAT 323 – Numerical Analysis I
 - MAT 110 – Mathematics for IT
 - BIT 311/MAT 322 – Operation Research I
 - MAT 304 – Complex Analysis
 - MAT 321 – Ordinary Differential Equations I
 - MAT 423 – Ordinary Differential Equations II
 - MAT 421 – Partial Differential Equations I
 - MAT 422 – Partial Differential Equations II
 - MAT 100 – General Mathematics
- Participate in setting, moderating, marking and analyzing examinations for undergraduate courses
- Supervise undergraduate students in Applied Mathematics projects
- B.Sc. Mathematics students' academic advisor

2018–May 2020 **Part-time Lecturer of Mathematics, Kibabii University, Bungoma, Kenya**

Responsibilities:

- Taught undergraduate students in the department of Mathematics
- Courses taught:
 - MAT 428 – Mathematical Modeling
 - MAT 323 – Numerical Analysis I
 - MAT 110 – Mathematics for IT
 - BIT 311/MAT 322 – Operation Research I
 - MAT 304 – Complex Analysis
 - MAT 203 – Vector Analysis
- Participated in setting, moderating, marking and analyzing examinations for undergraduate courses
- Supervised undergraduate students in Applied Mathematics projects
- B.Sc. Mathematics students' academic advisor

Feb 2017–Present **Mathematics Teacher & Head of Department, St. Francis Xavier Secondary School Shipalo, Kakamega, Kenya**

Responsibilities:

- Teaching Mathematics and Physics from Form 1-4
- Setting, moderating, marking and analyzing examinations
- Organizing and supervising laboratory practical lessons
- Offering guidance and counseling to students
- In charge of specific weeks within the term
- As Head of Mathematics and Science Department:
 - Making termly and yearly academic calendars
 - Supervising/monitoring effective curriculum delivery
 - Maintaining records of work covered books
 - Organizing inter-stream and inter-class competitions in mathematics
 - Organizing motivational talks

Research Supervision

2025 **Postgraduate Research Supervisor, Masinde Muliro University of Science and Technology**

Student: Nicholas Okuku Alele

Thesis: Modeling the Spread of Microplastic Pollutants in Aquatic Systems Using Advection-Diffusion Equation -**Ongoing**

Publications

- [1] **Owade Kennedy Jakob & Colletta Akinyi.** (2025). *Dynamics of the Fractional-Order Between-Host Cholera Model With Temperature-Dependence*. Abstract and Applied Analysis - **In Production (Wiley Online Library)**.
- [2] **Owade, K. J., Okaka, A., & Tireito, F.** (2024). *A Between-Host Cholera Mathematical Model Incorporating Temperature Dependence*. African Journal of Empirical Research.
- [3] **Owade Kennedy Jakob, Okaka Akinyi, & Frankline Tireito.** (2023). *A Mathematical Model on the Dynamics of In-Host Infection Cholera Disease with Vaccination*. Discrete Dynamics in Nature and Society.
- [4] **Jackob, K. O.** (2020). *Determination of the Equilibrium Point According to a Temporal Model of Rehydration in a Population with Cholera in Logistic Growth*. Journal of Advances in Mathematics and Computer Science.
- [5] **Nathan, O. M., & Jackob, K. O.** (2020). *Mathematical Modeling and Analysis of Mathematics Anxiety Behavior on Mathematics Performance in Kenya*. Journal of Advances in Mathematics and Computer Science.
- [6] **Nathan, O. M., & Jackob, K. O.** (2019). *Stability Analysis in a Mathematical Model of Corruption in Kenya*. Asian Research Journal of Mathematics.
- [7] **Owade Kennedy Jakob, Akinyi Colleta, & Owuor George Lawi.** (2019). *Modeling the Role of Rehydration and Antibiotic Treatment on Reduction of Cholera Mortality*. IOSR Journal of Mathematics.

Awards & Recognition

- 2025 **Certificate of Excellence in Reviewing, Vacunas - Elsevier**
- 2025 **Certificate of Excellence in Reviewing, Asian Journal of Research in Infectious Diseases**
- 2025 **Certificate of Excellence in Reviewing, Asian Research Journal of Mathematics**
- 2025 **Certificate of Excellence in Reviewing, Journal of Advances in Microbiology**
- 2012/2013 **Certificate of Completion of Peer Education Training, I Choose Life Africa, Kenya**

Professional Development

2022 **Competency Based Curriculum Training, Teachers Service Commission, St. Agnes Shibuye Girls**

2015/2016 **Graduate Internship**, *Association of African Universities*, Masinde Muliro University
Received AAU graduate internship grant of \$1,000 for 3 months

Community Service

2020–Present **Secretary and Head of Linkages**, *BodaBoda Medical Ventures Ltd*

- Provide structural management, coordination, and research expertise
- Develop organizational strategy and stakeholder linkages
- Support youth empowerment in the BodaBoda industry

2010–Present **Peer Educator**, *Various Institutions*

- Provide peer counseling for students and colleagues
- Develop and implement effective learning strategies
- Facilitate goal-setting and problem-solving sessions

Technical Skills

Programming **MATLAB** (Advanced), **Python** (Advanced), **R** (Advanced)

Mathematical **Mathematica** (Intermediate)
Software

Technical Writing **L^AT_EX** (Advanced)

Office Tools **Spreadsheet** (Advanced)

Languages

English Fluent

Swahili Fluent

Luo Fluent

Referees

- **Dr. David Angwenyi**
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Masinde Muliro University of Science and Technology
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- **Dr. Akinyi Colleta**
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- **Dr. Frankline Tireito**
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Declaration

I, the undersigned, certify that the information provided correctly describes my qualifications, experience, and personal details.

***Last updated on November 7, 2025**